



视觉与人工智能处理应用工作坊

Vision and Artificial Intelligence Processing Application

主办方
东姑阿都拉曼管理及工艺大学
ADVANCE RETRO SOLUTION SDN BHD (ARS)

宁夏职业技术大学
NINGXIA VOCATIONAL AND TECHNICAL UNIVERSITY



宁夏开放大学
NINGXIA OPEN UNIVERSITY

参与方



Real-Time AI Motion Tracking Workshop

实时 AI 运动跟踪工作坊

Building digital overlays from pixels to neural networks.
从像素到神经网络构建数字叠加层。

The mission is to track movement and classify objects

任务是跟踪运动并对物体进行分类

1. Connect Cloud to Camera

1. 了解如何将网络摄像头连接到云环境。

2. Process Images with OpenCV

2. 学习 OpenCV 图像处理的基础知识。

3. Apply Custom AI Models

3. 将自定义 AI 模型应用于实时视频。





Bridging the gap between the Python cloud and your local camera

跨越 Python 云端与本地摄像头之间的鸿沟

Browser
浏览器



`eval_js()`

Base64 Data
Base64 数据



Local Webcam (JS)
本地摄像头 (JS)

Google Colab
云端环境



NumPy Array
NumPy 数组

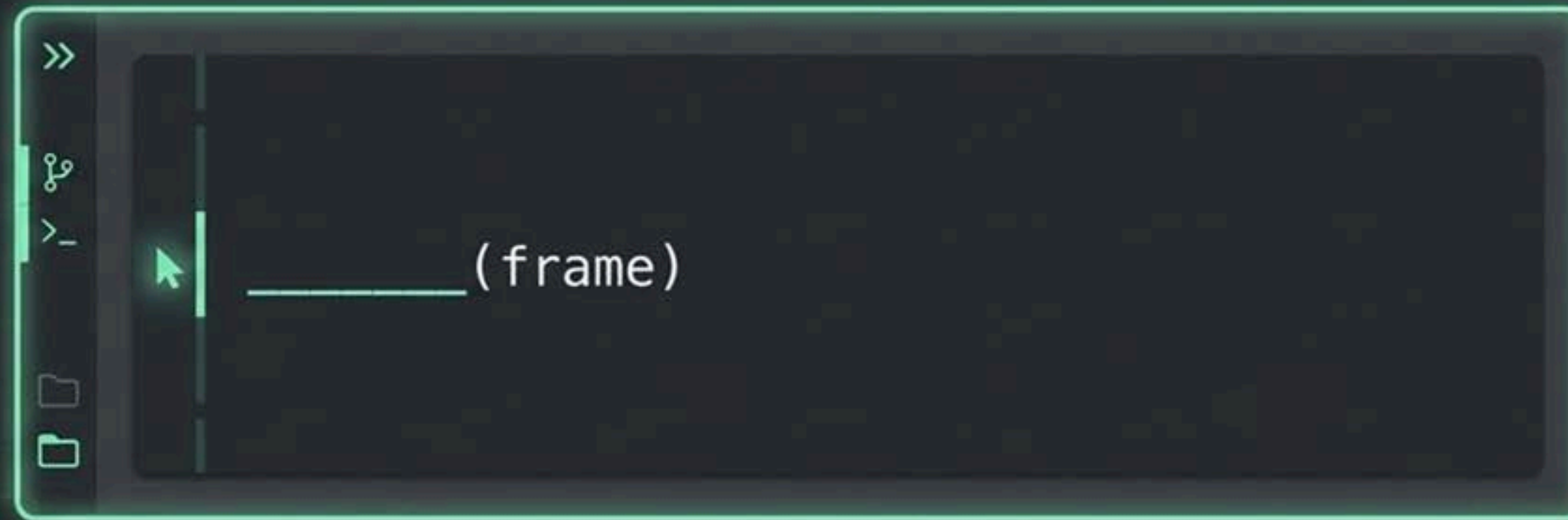


STUDENT TASK 1: Display the captured frame

学生任务 1: 显示捕获的帧

We successfully captured the image from the webcam. Now, fill in the blank to display it using the custom Colab function.

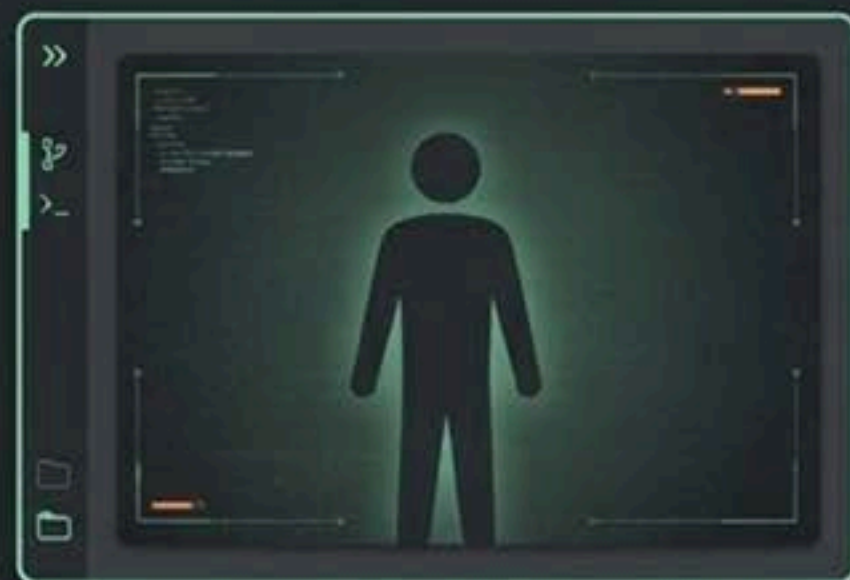
我们成功从网络摄像头捕获了图像。现在，填空以使用自定义的 Colab 函数显示它。



Hint: Use cv2_imshow
提示: 使用 cv2_imshow

Teaching the computer to see movement through pixel subtraction

通过像素相减让计算机看到运动



Previous Frame

上一帧



Current Frame

当前帧



Frame Difference

帧差 (cv2.absdiff)

OpenCV compares pixels over time. Moving objects become pure white;
static backgrounds become black.

OpenCV 随时间比较像素。移动的物体变成纯白色；静态背景变成黑色。



STUDENT TASKS: Complete the OpenCV motion tracking pipeline

学生任务：完成 OpenCV 运动跟踪管道

```
>> Convert to grayscale to simplify calculations.  
>> 转换为灰度图像以简化计算。  
>>  
gray = cv2.____(img, cv2.COLOR_BGR2GRAY)
```

Hint: cvtColor

```
>> Calculate the absolute difference.  
>> 计算绝对差异。  
>>  
frame_difference = cv2.____(previous_frame, gray)
```

Hint: absdiff

```
>> Apply a threshold to turn moving objects white.  
>> 应用阈值使移动物体变白。  
>>  
thresh = cv2.____(frame_difference, 25, 255, cv2.THRESH_BINARY)[1]
```

Hint: threshold

```
>> Draw a green rectangle on the overlay.  
>> 在叠加层上绘制绿色矩形。  
>>  
cv2.____(overlay, (x, y), (x+w, y+h), (0, 255, 0, 255), 3)
```

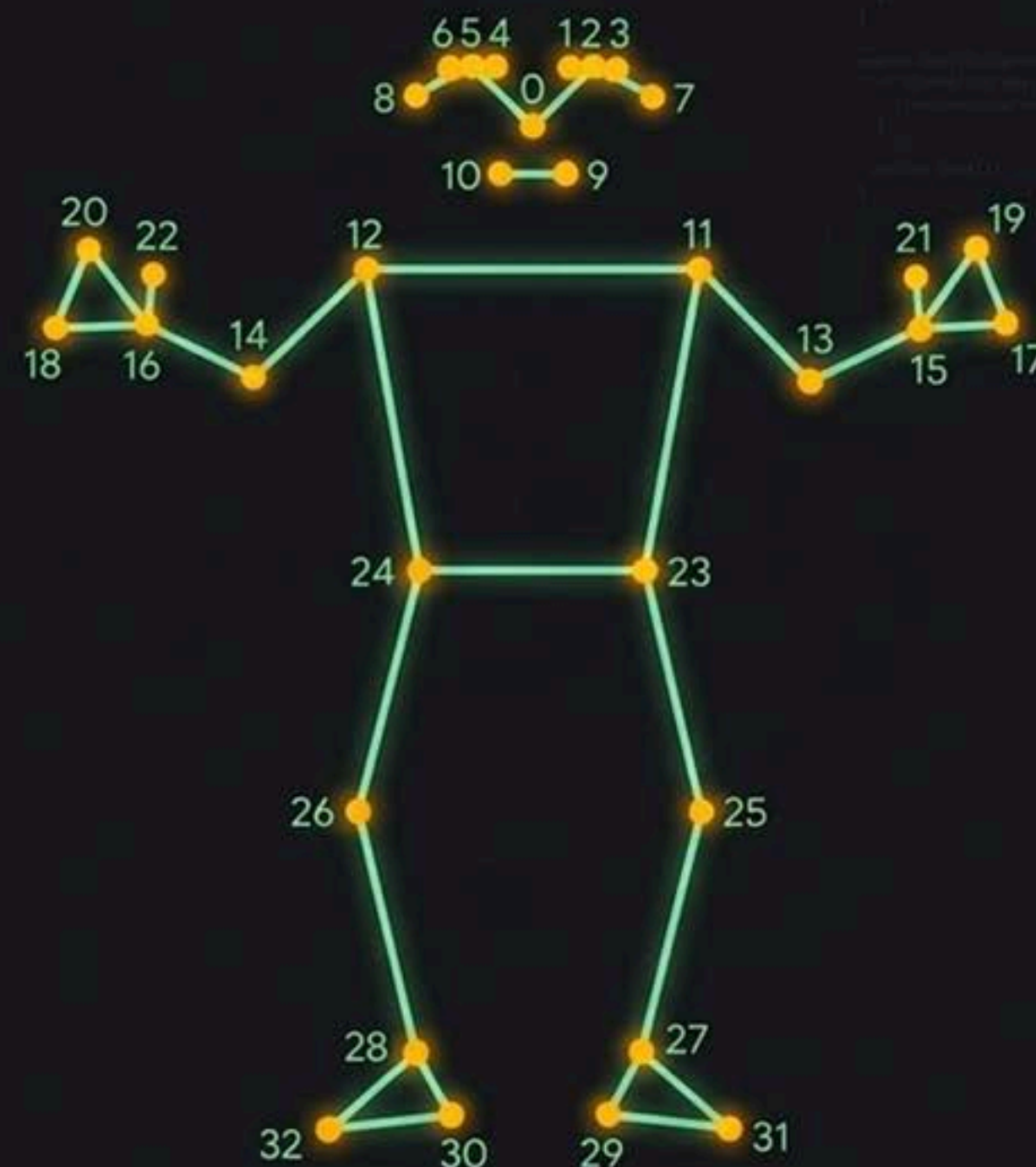
Hint: rectangle

Mapping the human body using 33 AI keypoints

使用 33 个 AI 关键点映射人体

Instead of just looking for moving pixels, the MediaPipe model uses neural networks to instantly identify 33 specific joints on the human body, regardless of the background.

MediaPipe 模型不再仅仅寻找移动的像素，而是使用神经网络瞬间识别出人体上特定的 33 个关节，不受背景影响。





STUDENT TASKS: Draw the skeleton overlay

学生任务：绘制骨架叠加层

>> Convert BGR to RGB for MediaPipe.
>> 将 BGR 转换为 RGB 以供 MediaPipe 使用。

```
img_rgb = cv2.__(img, cv2.COLOR_BGR2RGB)
```

Hint: cvtColor

>> Draw a circle for each joint (Yellow Dots).
>> 为每个关节画一个圆（黄点）。

```
cv2.__(overlay, (x, y), 5, (0, 255, 255, 255), -1)
```

Hint: circle

>> Draw a line to connect the joints (Green Lines).
>> 画一条线连接关节（绿线）。

```
cv2.__(overlay, start_point, end_point, (0, 255, 0, 255), 3)
```

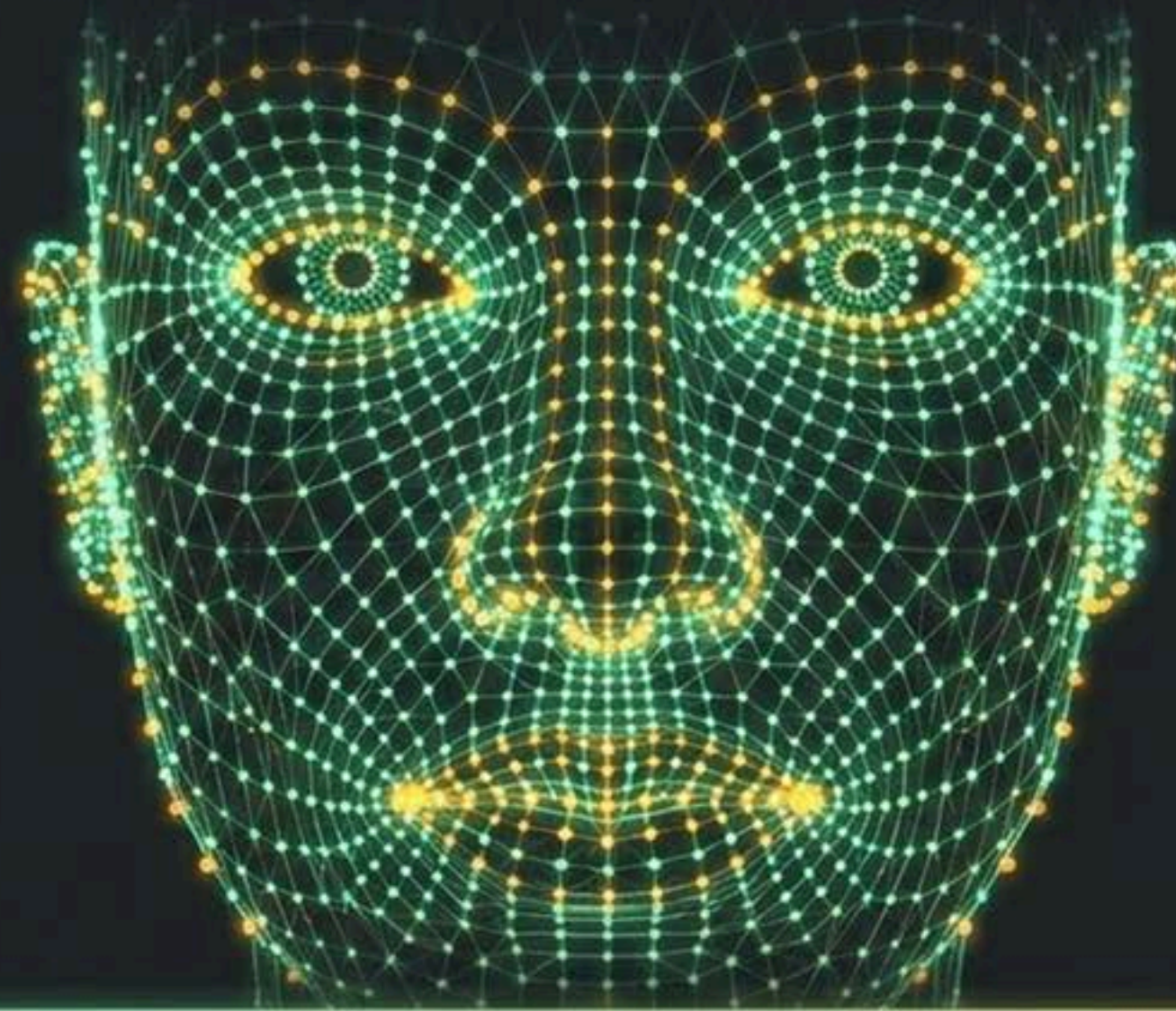
Hint: line

>> Put text on the screen to show success.
>> 在屏幕上放置文字以显示成功。

```
cv2.__(overlay, 'Skeleton Detected!', (20,40), ...)
```

Hint: putText

Real-time detection of 468 precise facial landmarks 实时检测 468 个精确的面部特征点



The pipeline remains exactly the same, but the density of the data array increases drastically.
管道保持完全相同，但数据数组的密度急剧增加。



STUDENT TASKS: Activate the face tracker

学生任务：激活面部跟踪器

Convert BGR to RGB. 将 BGR 转换为 RGB。

```
img_rgb = cv2.____(img, cv2.COLOR_BGR2RGB)
```

Hint: cvtColor

Draw a dot for every point on the face (468 loops). 为脸上的每个点画一个圆点（循环 468 次）。

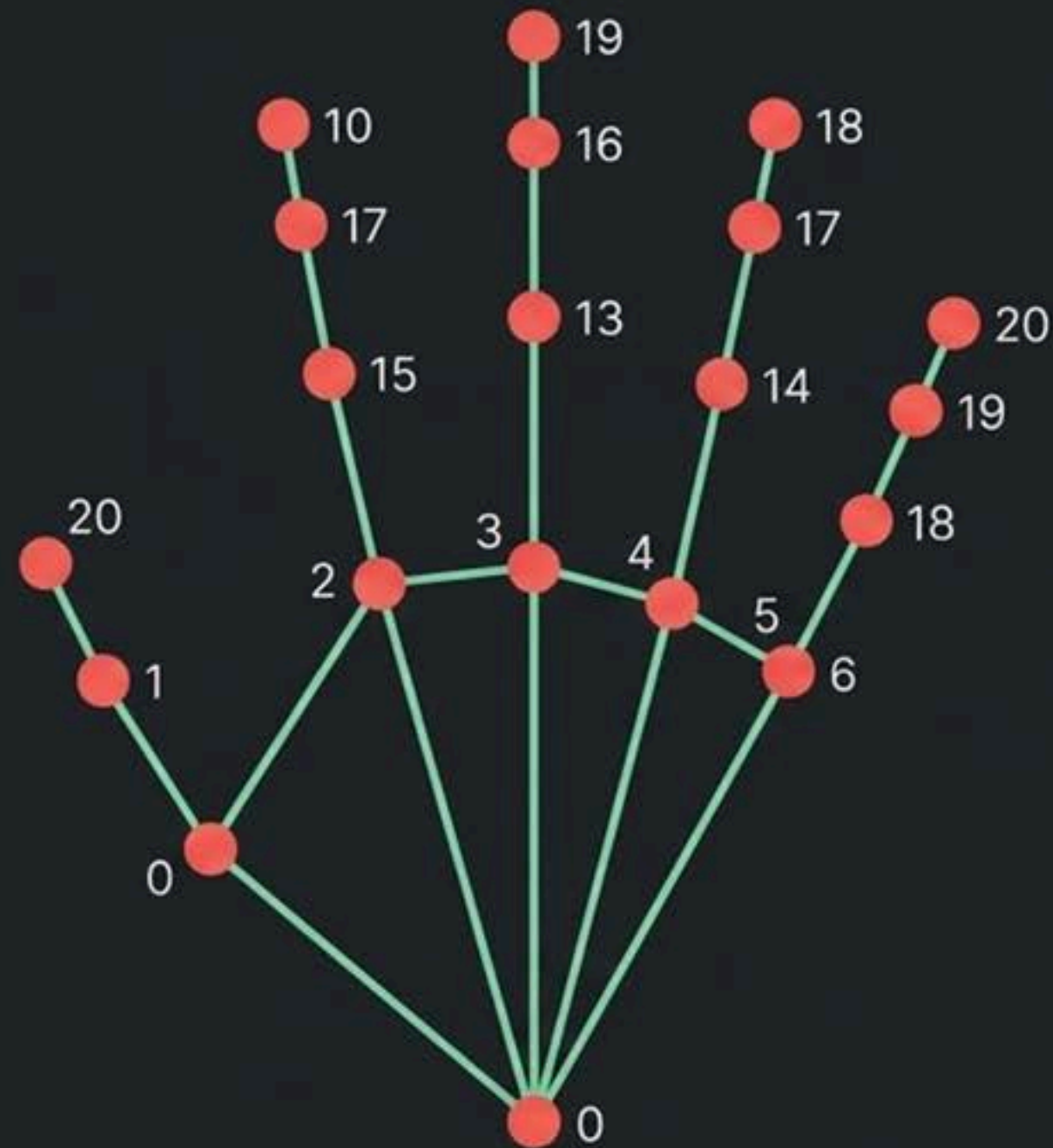
```
cv2.____(overlay, (x, y), 1, (255, 255, 0, 255), -1)
```

Hint: circle

Put text on the screen. 在屏幕上放置文字。

```
cv2.____(overlay, 'Face Tracked!', ...)
```

Hint: putText



Tracking 21 independent hand joints simultaneously 同时跟踪 21 个独立的手部关节

The AI categorizes the hand into 5 distinct fingers, tracking the wrist, MCP, PIP, DIP, and fingertips separately.

AI 将手分为 5 根不同的手指，分别跟踪手腕、掌指关节、近端指间关节、远端指间关节和指尖。



STUDENT TASKS: Connect the hand nodes

学生任务：连接手部节点

Draw a dot for every joint on the hand (Red Dots). 为手上的每个关节画一个圆点（红点）。

```
cv2.__(overlay, (x, y), 5, (0, 0, 255, 255), -1)
```

Hint: circle

Draw lines to connect the bones. 画线连接骨骼。

```
cv2.__(overlay, start_point, end_point, (255, 255, 0, 255), 3)
```

Hint: line

Add success text. 添加成功文本。

```
cv2.__(overlay, 'Hands Detected!', ...)
```

Hint: putText



Deploying a bespoke AI model for custom classification

部署定制的 AI 模型进行自定义分类



Unlike MediaPipe which looks for specific human traits, Teachable Machine uses a Convolutional Neural Network (CNN) to guess what the entire image represents based on your unique training data.

与寻找特定人类特征的 MediaPipe 不同，Teachable Machine 使用卷积神经网络 (CNN)，根据您独特的训练数据猜测整个图像代表什么。

STUDENT TASKS: Process the image and predict the class

学生任务：处理图像并预测类别

Resize a copy of the image to exactly 224x224 pixels.
将图像副本的大小精确调整为 224x224 像素。

```
model_image = cv2.____(img, (224, 224), interpolation=cv2.INTER_AREA)
```

Hint: resize

Find the index position of the highest prediction score.
找到最高预测分数的索引位置。

```
index = np.____(prediction)
```

Hint: argmax

Format and draw the text on the screen.
格式化并在屏幕上绘制文本。

```
cv2.____(overlay, display_text, ...)
```

Hint: putText

The Computer Vision Evolution Matrix

计算机视觉演进矩阵

Feature	OpenCV Motion	MediaPipe Landmarks	Teachable Machine
Method 方法	Pixel Difference 像素差异	AI Keypoint Mapping AI 关键点映射	Custom Classification 自定义分类
Object Awareness 物体感知	Doesn't know what is moving 不知道移动的是什么	Knows where human parts are 知道人体部位在哪里	Knows what the object is 知道物体是什么
Processing Power 处理能力	Low 低	Medium 中	High 高



The universal pipeline behind all AI overlays

所有 AI 叠加层背后的通用管道



Capture Pixels / 捕获像素

Capture Pixels / 捕获像素

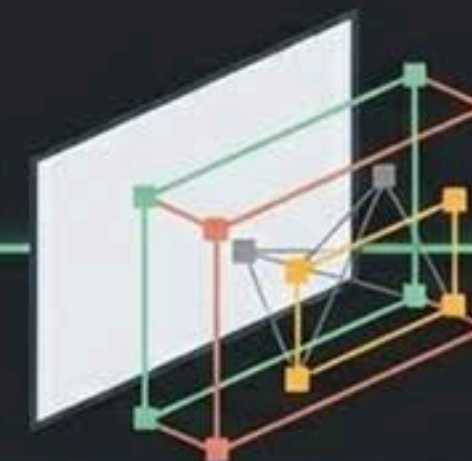
Pulling frames from the webcam to the cloud.
从网络摄像头拉取帧到云端。



Apply Math / 应用数学

Apply Math / 应用数学

Filtering pixels, mapping AI arrays, or predicting classes.
过滤像素、映射 AI 数组或预测类别。



Draw Overlay / 绘制叠加层

Using OpenCV shapes to visualize the data over the image.

使用 OpenCV 形状在图像上可视化数据。

AI vision is essentially complex mathematics layered cleanly over raw pixels.
AI 视觉本质上是干净地叠加在原始像素上的复杂数学。



TARUMT
TUNKU ABDUL RAHMAN
UNIVERSITY OF
MANAGEMENT AND TECHNOLOGY



Advance Retro Solution Sdn Bhd



亚 龙
YALONG

宁夏职业技术大学
NINGXIA VOCATIONAL AND TECHNICAL UNIVERSITY



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